

# ARIEF F-SA WIJAYA

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## PROFILE SUMMARY

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Fourth-semester Diploma (D3) student in Informatics Engineering at Bandung State Polytechnic with a strong interest in software development and structured solution design. Experienced in thoroughly analyzing problems, designing clear solution architectures, and managing complexity prior to implementation. Has a background in competitive programming that fosters a systematic, analytical, and efficiency-oriented mindset, enabling the development of modular, readable, and maintainable code. Prioritizes architectural quality, code readability, and long-term maintainability in every software solution.

## TECHNICAL SKILLS

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### Bahasa Pemrograman

C++, Python, JavaScript, PHP, Java

### Web Application Development

- Building well-structured web applications
- Designing frontend–backend data flows using API- and HTTP-based communication
- Managing application architecture using the MVC approach

### Software Engineering

- Problem solving and solution design based on requirements analysis
- Writing modular, readable, and maintainable code
- Debugging and refactoring

### Algorithms & Data Structures

- Advanced Data Structures, Dynamic Programming, Graph Algorithms, Binary Search
- Time and space complexity analysis

## EDUCATION

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**Politeknik Negeri Bandung** (*Bandung State Polytechnic*)

2024-Present

Diploma (D3) Program in Informatics Engineering

**SMAN 1 Subang** (*Senior High School 1 Subang*)

2021-2024

Math and Natural Science

## PROJECT EXPERIENCE

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### **Aplikasi Koperasi Syariah Terintegrasi**

*Team Project – Fullstack Developer*

Laravel, Vue.js, JavaScript, PHP

- Developed an integrated Sharia cooperative application with a complex business domain
- Applied feature-based, end-to-end (fullstack) task distribution instead of separating frontend and backend responsibilities
- Implemented a multi-role system with centralized access control
- Handled business workflows including savings, murabahah-based installment financing, POS cash transactions, and inventory management
- Designed data structures and application flows to ensure consistency between financial and inventory modules

### **Productivity Application (Java Desktop)**

*Team Project – Lead Developer*

Java, JavaFX, PostgreSQL, SQLite

- Designed a desktop application using a layered MVC architecture
- Applied multiple design patterns to maintain code modularity and scalability
- Implemented a PostgreSQL database with SQLite fallback stored in the default user directory across operating systems
- Built a cross-platform, installable application, including Java runtime configuration
- Developed comprehensive unit testing and structured input–output validation
- Designed UI/UX with a focus on readability, usage flow, and overall user experience

### **Block Shooter Game**

*Team Project – Lead Developer*

C, Raylib

- Led the development of a simple 2D game using C and raylib
- Manually implemented low-level data structures
- Designed the game loop, object behavior, and game state persistence
- Managed user experience considerations and team collaboration throughout development

## ACHIEVEMENTS

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| <b>ACM/ICPC Regional Asia</b><br><i>Honourable Mention</i>                                                                             | 2025 |
| <b>ACM/ICPC Provincial – Indonesia National Contest</b><br><i>Honourable Mention; Qualified for the ACM/ICPC Asia Regional Contest</i> | 2025 |
| <b>Olimpiade Sains Nasional bidang Komputer/Informatika</b><br><i>District Champion; Qualified for the Provincial Level</i>            | 2022 |

## ORGANIZATIONAL EXPERIENCE

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| <b>Junior Staff Member – Computer Science Student Association</b><br>Politeknik Negeri Bandung ( <i>Bandung State Polytechnic</i> )                                                                                                                                                                                                                                                                                                                            | 2025-sekarang |
| <ul style="list-style-type: none"><li>• Assisted in planning and executing academic knowledge-sharing programs across study programs within the Informatics Engineering Department</li><li>• Involved in managing bootcamp activities and competition training programs, including participant management and academic coordination</li><li>• Supported the preparation and execution of student competitions, both administratively and technically</li></ul> |               |
| <b>Head of Programming Division – School Computer Club</b><br>SMAN 1 Subang ( <i>Senior High School 1 Subang</i> )                                                                                                                                                                                                                                                                                                                                             | 2022-2023     |
| <ul style="list-style-type: none"><li>• Coordinated programming activities and collaborative projects among club members</li><li>• Designed and conducted training programs ranging from basic to advanced programming</li><li>• Mentored members in project development and problem-solving skill improvement</li><li>• Presented project outcomes at school events to encourage collaboration and innovation</li></ul>                                       |               |